

BIOFORM AI

Physics-Informed Neural Networks (PINNs) for Real-Time Biomechanical Fatigue & Form Tracking

Final Project Report

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Academic Year 2026

Note: Rehabilitation functionality is intended for movement-monitoring support only and is not a medical diagnostic system.

Executive Summary

BioForm AI is a web-based biomechanical monitoring prototype that combines computer vision, exercise-specific motion analysis and a Physics-Informed Neural Network (PINN) to estimate fatigue and assess exercise form from ordinary RGB video. The current implementation supports 33 configured gym and rehabilitation-support movements, video upload and browser-camera recording, joint-angle extraction, range of motion, movement velocity, bilateral symmetry, repetition counting, form scoring, session-specific feedback, fatigue estimation, model training and evaluation, and automatic result graphs.

The software is designed as an academic prototype. Its current supervised fatigue target is a biomechanical proxy; optional RPE is stored as metadata and is not silently treated as clinical ground truth. Therefore, model metrics describe performance against the current target, not clinical fatigue accuracy.

Project Status

Module	Status	Remarks
Computer vision backend	Completed	MediaPipe/OpenCV pipeline implemented
Biomechanical features	Completed	Angles, ROM, velocity, symmetry, reps
PINN model	Completed	Data loss + physics loss, training, model saving
Web frontend	Completed	Flask website with upload/camera/session pages
Per-video graphs	Completed	Fatigue, form, ROM, symmetry, velocity
Model evaluation	Completed in code	MAE, MSE, RMSE and R^2 generated after training
Real-world validation	Pending data collection	Requires multiple users/videos and held-out labels
Clinical validation	Not claimed	Outside current academic prototype scope

1. Introduction

Fatigue and poor movement form can alter exercise performance and may increase movement-related risk. Conventional monitoring can require coaches, wearable sensors or laboratory equipment. BioForm AI investigates a low-cost alternative in which a standard RGB camera is used to estimate pose landmarks, derive biomechanical features, and provide immediate session-level analysis.

The central research idea is to combine data-driven learning with a simplified fatigue-dynamics constraint. Instead of allowing the neural network to learn only from target values, the PINN also

penalizes predictions that violate a selected physical relationship between fatigue, movement effort and recovery.

2. Problem Statement

There is a need for accessible exercise monitoring that can quantify movement quality and fatigue without requiring specialized wearable hardware. The project addresses this by building an integrated software pipeline that accepts exercise video, extracts pose and biomechanical features, estimates fatigue, evaluates form, and visualizes results in a web interface.

3. Objectives

- Detect human pose from uploaded or browser-recorded exercise video.
- Calculate joint angles, range of motion, velocity and left-right symmetry.
- Count repetitions or static-hold duration for configured exercises.
- Provide exercise-specific form scores and corrective feedback.
- Train and apply a Physics-Informed Neural Network for fatigue estimation.
- Generate per-session summaries and video-specific graphs.
- Evaluate the regression model using MAE, MSE, RMSE and R^2 .
- Provide a usable local web interface for demonstration and testing.

4. System Architecture

Video / Browser Camera	MediaPipe Pose	Biomechanical Features	Exercise & Form Logic	PINN Fatigue Model	Web Results & Graphs
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The browser sends a recorded or uploaded video to the Flask backend. The backend runs pose estimation frame by frame, derives biomechanical variables, applies exercise-specific state logic, and optionally loads the trained PINN model. Session data, annotated video, summary JSON and graphs are saved in a unique session folder and rendered back to the user.

5. Biomechanical Feature Extraction

Pose landmarks are used to form three-point joint geometries. For vectors BA and BC around a joint B, the included angle is computed using the cosine relationship:

$$\theta = \cos^{-1}[(\mathbf{BA} \cdot \mathbf{BC}) / (|\mathbf{BA}| |\mathbf{BC}|)]$$

The implemented feature vector contains left/right knee, hip, elbow, shoulder and ankle angles; knee and hip ROM; movement velocity; symmetry difference; normalized movement effort; repetition count; and normalized time.

6. Exercise Analysis and Form Tracking

Each configured exercise defines a primary joint, low/high motion thresholds, phase names, repetition trigger and recommended camera view. A state machine counts a repetition only after the movement transitions through the expected phases. Static exercises such as planks use hold duration instead of rep counting.

Form scoring begins from a general symmetry/control score and then applies exercise-specific checks. Examples include squat/lunge depth, upper-arm stability during curls, elbow position during raises, trunk alignment in push-ups and planks, and more conservative symmetry/control rules in rehabilitation-support mode. These thresholds remain prototype rules and should be empirically calibrated using representative recordings.

7. Physics-Informed Neural Network

The fatigue predictor is a fully connected neural network with Tanh hidden activations and a sigmoid output. Two positive trainable parameters, α and β , are represented through exponentiated log-parameters. The simplified fatigue dynamics used by the prototype are:

$$dF/dt = \alpha E(t) - \beta F(t)$$

where $F(t)$ is normalized fatigue and $E(t)$ is normalized movement effort. The physics residual is the difference between the model-derived time gradient and the right-hand side of this equation.

Training minimizes:

$$L_{\text{total}} = L_{\text{data}} + \lambda L_{\text{physics}}$$

The implementation uses a physics weight $\lambda = 0.20$. A session-level GroupShuffleSplit is used when two or more sessions exist so frames from the same session are not intentionally mixed between training and testing.

8. Web Application

- Home dashboard with project/model status.
- Analyze page with Gym/Rehabilitation selection and exercise dropdown.
- Video upload and browser-camera recording.
- Optional RPE metadata collection.
- Session result page with annotated video, metrics and graphs.
- Training page for PINN training.
- Metrics page showing regression evaluation.
- Calibration page for editing exercise thresholds.
- Session history page for reviewing prior analyses.

The website runs locally using Flask. Browser-camera recording uses `getUserMedia/MediaRecorder` and sends the completed clip to the same analysis pipeline. This is not a low-latency live stream; it is a browser-recorded session analyzed after capture.

9. Result Generation

For every analyzed session, the application creates a unique folder containing `session.csv`, `summary.json`, `annotated.mp4` and graphs for fatigue, form score, ROM, symmetry and movement velocity. Model training produces a latest checkpoint, a timestamped versioned checkpoint, training-loss graph, `evaluation.csv`, actual-vs-predicted line plot and scatter plot.

10. Model Evaluation

Because fatigue is predicted as a continuous value, the project uses regression metrics rather than classification accuracy. MAE expresses the average absolute prediction error; MSE penalizes large errors more strongly; RMSE returns the squared-error measure to the target scale; and R^2 describes explained variance relative to the test-set mean baseline.

A previous prototype training run reported $MSE = 0.000902$ and $MAE = 0.019473$ (approximately 1.95 percentage points). These figures should be treated as previous-run results against the proxy target. The final web version is designed to regenerate MAE, MSE, RMSE and R^2 after training on the final collected sessions.

11. Testing and Validation Plan

Code-level tests verify angle calculation, repetition-state transitions and form scoring. Final experimental validation requires real exercise recordings. At minimum, the project should test Squat, Bicep Curl, Lunge, Push Up and Shoulder Press across multiple sessions/users, manually verify rep counts, inspect representative joint-angle frames and evaluate the trained model on held-out sessions.

For stronger fatigue validity, collect an external target such as session RPE, expert-labelled fatigue or physiological measurement. The current web application stores RPE as metadata but deliberately does not treat it as clinical ground truth.

12. Limitations

- Single-camera pose estimates are affected by occlusion, camera angle and landmark visibility.
- Exercise thresholds are prototype values and require empirical calibration.
- The current fatigue target is a biomechanical proxy, not a physiological gold standard.
- The system has not undergone clinical validation.

- Browser camera mode records and then analyzes a clip; it is not a production real-time streaming service.

13. Future Work

- Collect larger multi-subject datasets with validated fatigue labels.
- Add automatic exercise recognition using a separately labelled classifier.
- Add subject-specific calibration and longitudinal progress tracking.
- Explore multi-view pose estimation or depth cameras for occlusion robustness.
- Evaluate the system with clinicians/coaches for rehabilitation and sports use cases.
- Deploy the web application to a secure server with authenticated user accounts if required.

14. Conclusion

BioForm AI demonstrates an end-to-end academic prototype that combines computer vision, biomechanical feature extraction, exercise-specific form analysis and physics-informed learning in a web application. The core software implementation is complete. The remaining scientific work is empirical validation: collecting representative sessions, calibrating exercise rules, obtaining stronger fatigue labels and reporting final held-out metrics without overstating clinical accuracy.

Appendix A — Final Demo Flow

- Open the BioForm AI website.
- Select Gym or Rehabilitation and choose an exercise.
- Upload or record a video.
- Show annotated output, repetition count, form score, ROM, symmetry and fatigue.
- Show the session-specific graphs.
- Train the PINN after collecting multiple sessions.
- Open the Metrics page and explain MAE, MSE, RMSE and R^2 .
- Explain limitations and future validation honestly.